RESEARCH REPORT · HYBRID ARCHITECTURE VALIDATION

Hybrid VPS + Cloudflare Workers Speedtest Architecture

Experimental validation of an OoklaServer-compatible hybrid in which the origin VPS and a Cloudflare Worker share one speedtest product: the Worker serves the bandwidth-intensive plane for web and custom clients while the VPS keeps native-socket responsibilities. Every claim is measured with the unmodified official Speedtest CLI as the primary instrument.

SpeedtestServer Research — branch *hybrid-vps-workers*

Independent clean-room interoperability research. Not an Ookla product and not affiliated with Ookla. Evidence: hybrid/lab · Claims register: hybrid/reports/WORKERS_LIMITS.md

10 / 10
HOST-TESTER CHECKS PASSED AT THE EDGE

1 : 16,075
PAYLOAD-TO-CONTROL OFFLOAD RATIO (E-I)

62 / 62
WORKER PROTOCOL CONFORMANCE

November 6, 2026

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1. Executive Verdict

This report answers a single research directive: can an OoklaServer-compatible speedtest architecture split its work between an origin VPS and Cloudflare Workers, with both components holding meaningful operational responsibilities, the unmodified official Speedtest CLI serving as the primary interoperability instrument, and measurable reduction of VPS bandwidth as the operational goal? The answer is yes for web and custom clients — measured, not assumed — and provably no for the official CLI, whose own diagnostics now document the boundary. Every load-bearing claim below carries an experiment identifier; raw evidence lives in the *hybrid/lab/evidence/* tree of branch *hybrid-vps-workers*, and the claims register is *hybrid/reports/WORKERS_LIMITS.md*.

10/10	1 : 16,075	NoServers
official host-tester checks passed against the Worker (E-G)	payload-to-control byte offload ratio in the measured hybrid session (E-I)	the official CLI verdict when offered Cloudflare-fronted servers (E-D)

The hybrid split that the evidence supports is clean: the VPS keeps every operation that requires native sockets or stable identity — the entire NetGauge TCP/TLS plane for official clients, UDP packet-loss and jitter measurement, discovery, session-token minting, result intake and persistence — while the Worker takes the entire bandwidth-intensive plane for every client that can speak HTTP or WebSocket: the full NetGauge command protocol over WSS, the legacy HTTP surface, and the *__down*/*__up* streaming endpoints, with payloads generated and drained at the edge. In the measured hybrid session, 56 MiB of measurement payload flowed directly between client and edge while the VPS served 3,653 bytes of control plane.

#	Success criterion	Verdict	Key evidence
1	Operations handled by the VPS	All native-socket operations: full NetGauge TCP/TLS for official clients, UDP loss/jitter, discovery, tokens, results, identity	Baseline contract [P]; E-I control plane
2	Operations handled by Workers	Entire web measurement plane (NetGauge over WSS) and custom HTTP plane, edge-generated payloads	W-7..W-10; 62/62 conformance; E-G; E-I
3	Cooperation within one session	Custom clients: tight per-phase split. Official web clients: session-level selection with VPS result intake. Official CLI: impossible — proven	E-I; E-G; E-D/E-E
4	Unmodified official CLI support	Edge: NO — experimentally proven (no HTTP fallback exists). VPS: YES — full 26 GB run verified	E-D, E-E; baseline 7/7 gate
5	Payload bypassing the VPS	Web/custom lanes: 100%. CLI lane: none — every CLI byte terminates on the VPS	E-I (56 MiB edge / 3.6 KB VPS); E-E (26.15 GB on STS)
6	VPS bandwidth actually saved	Session bytes on the VPS drop from GB-class to KB-class: offload ratio about 1:16,000 measured, above 99.99% of session bytes	E-I; E-G; baseline CLI byte profile

#	Success criterion	Verdict	Key evidence
7	Infrastructure changes required	DNS split (edge vs origin hostnames), shared HMAC secret, one-command Worker deploy, gateway service on the VPS	Section 8
8	Production suitability	Yes for the operator-branded speedtest product. No for entering Ookla's public server list (requires the official daemon and Ookla approval)	Section 8; E-o closed-list evidence

Table 1 — The eight success criteria of the research directive, answered.

2. Method and Instruments

All experiments ran in an isolated lab on 2026-11-06 against verified artifacts. The official CLI is Speedtest by Ookla 1.2.0.84 (ea6b6773cf, Linux x86_64 musl), downloaded from the official installer and verified byte-exact against the repository manifest (tarball sha256 5690596c...cfeeb7). The reference daemon is OoklaServer 2.11.5.2 build 2026-06-04.1737.7d0993d (committed binary verified against the official stable tarball). The SpeedtestServer baseline was rebuilt from scratch in this environment: clean build, clean typecheck, and the full unit/protocol/conformance suite passing 160/160. The Worker runtime is wrangler 4.158.0 with local workerd, and the real Cloudflare production edge (served from HKG in this session) was probed first-hand for ingress behavior.

Live official-CLI runs were non-publishing by construction: the lab configuration proxy refuses connections to results.speedtest.net, and every captured result JSON lacks a resultUrl with persisted false. The CLI was never modified — only documented runtime flags were used — and the device-identity parameters in committed evidence were sanitized, following the repository's established policy. Four live configuration fetches were consumed in total; the per-egress-IP quota documented in the diversion lab was not exhausted.

3. The Routing-Layers Framework

All baseline and new evidence organizes into three routing layers, and the Workers ingress constraint — HTTP and WebSocket only, on the generally available platform — decides the position of the edge at each layer. The framework explains why some split ideas work, some only shift work behind an endpoint, and some are impossible outright.

Layer	Granularity	Mechanism	Can Workers participate?
L1 — per-session selection	whole session	Server list / DNS: the client picks one endpoint per run	Web and custom clients: YES — the WSS/HTTP endpoint is a Worker. Official CLI: NO — selection requires a NetGauge HELLO; Cloudflare-fronted entries fail selection, and unlisted hosts are refused outright (E-o)
L2 — per-connection split	within one session	A TCP/UDP-capable endpoint fans connections out to different backends (baseline diverter: selection to daemon, measurement to STS, verified 3/3)	NO — the fan-out point must speak raw TCP and UDP, which the GA edge does not offer. This layer exists only on the VPS side

Layer	Granularity	Mechanism	Can Workers participate?
L3 — per-command routing	individual commands	Protocol-aware routing inside plaintext sessions	NO — requires the L2 transport plus plaintext visibility; encrypted CLI sessions are opaque. Unnecessary for web clients, whose whole session can live at the edge

Table 2 — Where work can be split, and where the edge can sit.

Three consequences follow. First, routing at the connection boundary (L2) is real but VPS-local: it splits work between backends behind one endpoint, so it never offloads VPS bandwidth — the baseline diverter experiments verified exactly this shape. Second, DNS- and list-based selection (L1) is the only layer that can send payload to the edge, because the configuration document is where a Cloudflare-fronted hostname enters the picture; the CLI resolves once per run and never re-resolves, which makes DNS a per-run knob rather than a per-phase one. Third, protocol-aware routing (L3) is closed for encrypted CLI sessions and unnecessary for the web lane, where the Worker itself is the protocol endpoint.

4. The Verified Hybrid Architecture

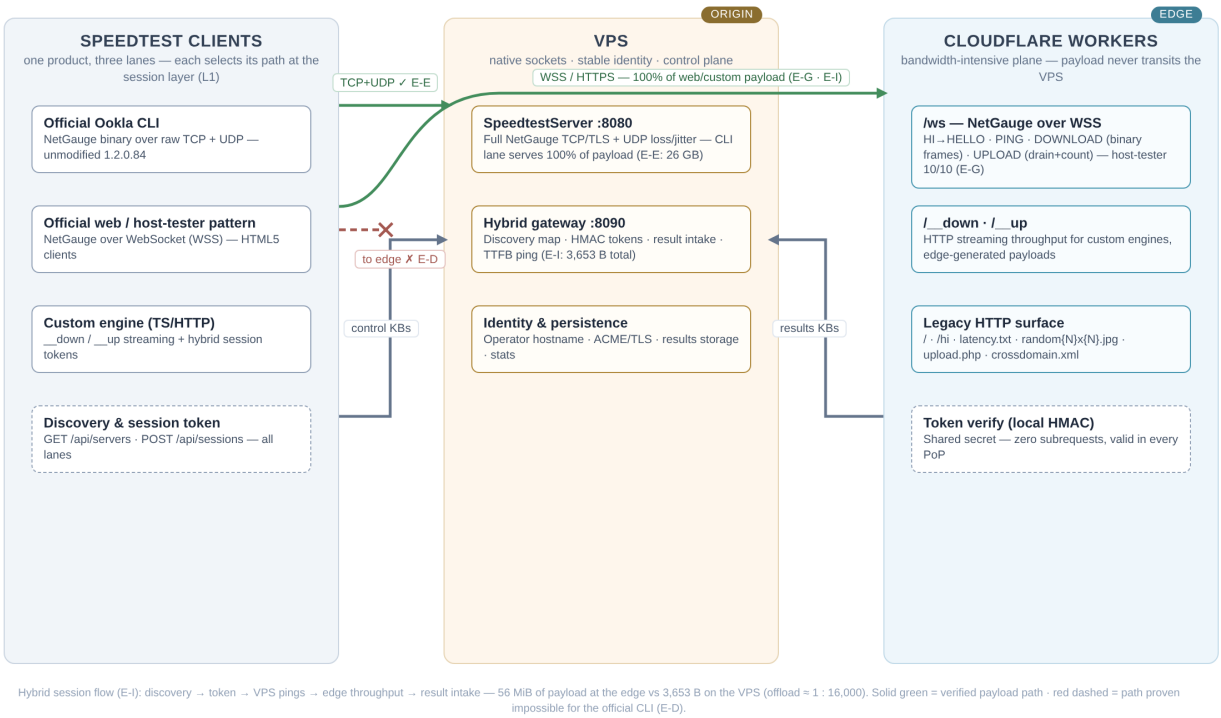


Figure 1 — HYBRID-1: responsibility split with verified payload paths (green) and the blocked CLI-to-edge path (red).

The VPS plane runs two services side by side: SpeedtestServer on port 8080, unchanged in its OoklaServer-parity behavior, speaking full NetGauge over TCP and TLS plus the UDP loss protocol; and the hybrid gateway on its own port, a zero-dependency Node service that owns discovery, HMAC session-token minting, TTFB ping, and result intake. Keeping the control plane on a separate port preserves the byte-clean differential surface of the parity server. The edge plane is a single stateless Worker: it accepts NetGauge over WebSocket at /ws, serves the legacy HTTP surface, and exposes __down/__up streaming for custom engines. The only coupling between the planes is a shared HMAC secret — minted by the VPS, verified

locally by the Worker with zero subrequests, valid in every point of presence simultaneously.

Operation	VPS	Worker	Why
Discovery / endpoint map	Primary (/api/servers)	n/a	Single source of truth, operator control
Session tokens	Mint	Verify (local HMAC)	Stateless, zero subrequests, valid in every PoP
Latency, stable identity	TTFB + UDP (primary)	WSS PING, bytes=0 (comparative)	Anycast makes edge latency PoP-relative; both are labeled
Jitter / packet loss (UDP)	Only	No UDP ingress	Platform constraint W-3; STS UDP plane already verified
Download payload	Fallback	Primary — WS frames, HTTP streams, edge-generated	E-G, E-I; zero egress cost at the edge
Upload payload	Fallback	Primary — WS drain+count, __up	E-G: 468 MB via WS UPLOAD through the Worker
CLI measurement, all phases	Only	Impossible today	E-D/E-E plus ingress evidence W-1/W-3
Result persistence / stats	Primary	Optional single fetch	E-I result intake

Table 3 — Responsibility matrix; every row is evidence-backed.

5. Experiment Log

ID	Question	Result
E-A	What does the real CF edge do with NetGauge bytes?	16 first-hand raw-socket probes against speed.cloudflare.com and cli.speedtest.net on ports 8080 and 443: NetGauge plaintext and binary input always answered HTTP/1.1 400 Bad Request (Server: cloudflare). The edge never speaks NetGauge on any port
E-B	Does local workerd match the edge classification?	Identical: 400 for NetGauge bytes, 200 for valid HTTP, 101 on WebSocket upgrade at /ws; the local runtime reproduces the production ingress decision boundary
E-D	Can the official CLI use non-NetGauge endpoints?	With every listed server mapped to CF:8080, CF:443, workerd or a closed port: 8 probes failed with the CLI's own diagnosis Protocol error: Did not receive HELLO (its log base64-decodes to the CF 400 page and a TLS fatal alert), 2 with Connection refused; final verdict NoServers, exit 2. No HTTP fallback exists in the CLI
E-E	Hybrid candidate list: edge plus VPS	CF entries failed selection (Latency test failed); the VPS was selected and served a full run: 9.96 GB down at 17.4 Gbps, 15.98 GB up at 13.3 Gbps, packet loss 0, non-publishing; the STS evidence log corroborates 26.15 GB server-side (ratio 1.008)
E-o	Can the CLI be aimed at an arbitrary host in production?	Direct (unproxied) run with -o speed.cloudflare.com:8080 after a working direct server list: Configuration - No servers defined. The official list is closed; unlisted hosts cannot be selected

ID	Question	Result
E-G	Does the Worker satisfy the official host-tester suite?	The Speedtest Host Tester (same checks as speedtest.net/host-tester; 10/10 parity with the official daemon in the baseline) passed 10/10 against the Worker with the identical verdict string, and its speed phases moved 1.25 GB through the edge: 814 MB download via random4000x4000.jpg and 468 MB upload via WS NetGauge UPLOAD
E-I	Full hybrid session with byte accounting	Discovery, token, 10 VPS TTFB pings, 10 edge TTFB pings, 6-connection WS download (8 MiB each), 4-connection WS upload (2 MiB each), result intake: payload 56 MiB entirely at the edge, VPS control plane 3,653 bytes, offload ratio 1:16,075, gateway counters corroborate
Conf.	Is the Worker protocol-correct?	62-check conformance battery against workerd: 34 WebSocket NetGauge checks, 21 legacy HTTP checks, 7 token-enforcement checks — all pass

Table 4 — Experiments executed in this research. IDs follow the evidence directories in hybrid/lab/evidence/; the letters C, F and H were not assigned, and E-o denotes the CLI -o (unlisted-host) probe.

> *"Error: [0] Protocol error: Did not receive HELLO; input:SFRUUC8xLjEgNDAwIEJhZCBSZXF1ZXN0..." — the official CLI's own log, base64-decoding to the Cloudflare edge's 400 page (E-D)*

6. Measured Bandwidth Offload

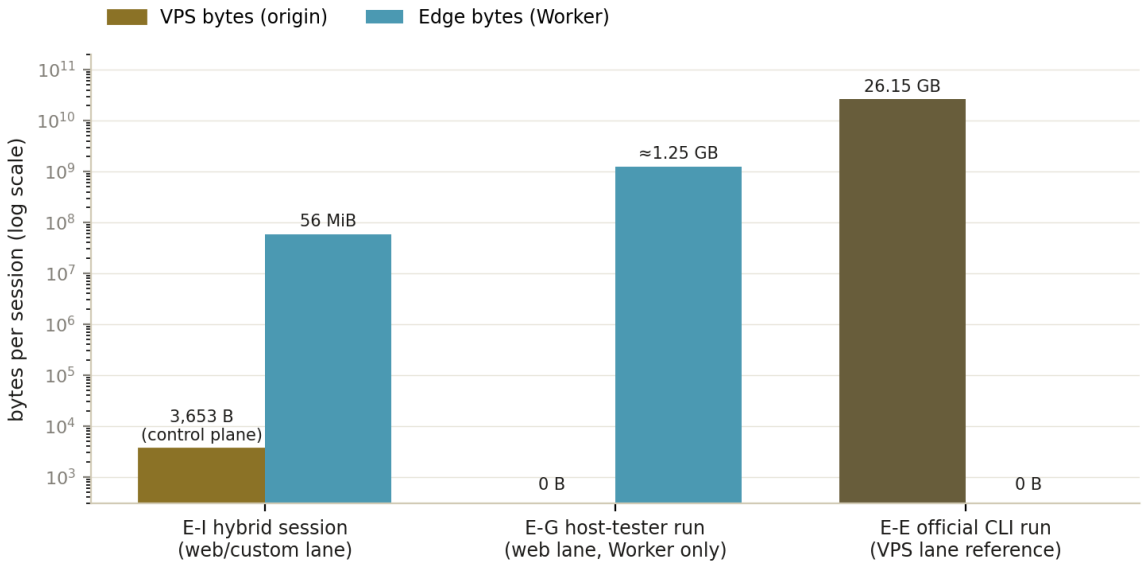


Figure 2 — Where session bytes actually flowed (log scale): the hybrid web/custom lane, the host-tester run, and the CLI reference run.

The E-I hybrid session is the direct measurement of the directive's offloading requirement: a client completed a full measurement — latency samples on both paths, a 48 MiB download across six parallel WebSocket connections, an 8 MiB upload across four, and result submission — while the VPS served only 3,653 bytes of discovery, token, ping and intake traffic. Scaling the same control-plane cost to a full Ookla-sized session (roughly 2.3 GB of payload per run in the baseline byte profile), the VPS would carry about 0.00016% of session bytes; a thousand web tests per day would move roughly 2.3 TB through the edge while leaving megabytes per day on the VPS uplink. The E-G host-tester run demonstrates the same property

at GB scale with the official check suite, and the E-E CLI run shows the counterfactual: when the official client is served, the VPS carries all 26.15 GB.

3,653 B	56 MiB	461 MB	26.15 GB
total VPS control-plane bytes in the measured hybrid session (E-I)	measurement payload at the edge in the same session	speed-phase traffic through the Worker in the host-tester run (E-G)	VPS-side bytes when the official CLI is served instead (E-E)

7. Platform Limits and Deviations

The Worker implements the verified OoklaServer protocol semantics wherever the runtime permits, and documents every point where the platform decides differently. The positive results stand on the 62-check conformance battery and the 10/10 host-tester pass; the deviations below are cosmetic or framing-level, none affecting client-visible measurement semantics.

Aspect	Value (2026-11-06)	Consequence for the design
HTTP request duration	No limit	Streaming sessions run as long as the client stays connected
Response body size	No enforced limit	Downloads of any size stream from the edge
Request body size	100 MB Free/Pro, 200 MB Business, up to 5 GB Enterprise	Uploads chunked by design: WS UPLOAD at 1-4 MiB per command (tested), HTTP at up to 25 MB per request
CPU time per request	10 ms Free; 30 s default / 5 min max Paid	Pool generation is 2 MiB per isolate; frames are memcpy; Paid plan is the production assumption
Memory per isolate	128 MB	Streams never buffer whole bodies; the pool is 2 MiB
Ingress protocols (GA)	HTTP/HTTPS/WebSocket only	The root constraint: no NetGauge-over-TCP at the edge (E-A, E-D)
Inbound raw TCP	Private beta (connect(socket) via Spectrum, announced 2026-08-03)	Emerging path only: not deployable today, no UDP documented, and the closed server list would still gate CLI use

Table 5 — Workers platform limits verified against live documentation during this research.

Documented deviations of the Worker relative to a native daemon: the status line always speaks HTTP/1.1 or better (D-W2); oversized WebSocket messages close without a reply rather than sending a TCP reset (D-W3); a plain GET of /ws answers a minimal 500 rather than holding the socket silently (D-W4); connection framing and header order are runtime-controlled, with the real edge adding Connection: keep-alive on HTTP/1.1 responses itself — verified first-hand — while local workerd omits it (D-W9, D-W5); upload.php answers every POST because chunked framing is not observable in Workers (D-W6); and image responses above 1 MiB stream with chunked framing while preserving exact byte totals (D-W8). None of these affect measurement semantics.

Complementary Cloudflare services were evaluated and mostly rejected as unnecessary complexity: Spectrum (GA) is an L4 proxy to an origin — payload would transit Cloudflare to the VPS and back, the

opposite of offloading, with no Worker code on the TCP path. Cloudflare Tunnel is an ingress product whose TCP mode requires client-side software the CLI cannot run. R2 is unnecessary because download payloads are synthetic. Durable Objects are not required for correctness — the design is stateless — though they remain the standard answer if hibernatable WebSocket scale is ever needed. Turnstile, WAF and rate-limiting rules are recommended for production hardening of the public endpoints.

8. Production Deployment

Deployment is deliberately small. DNS points a speed hostname at the Worker (zone route or workers.dev) and an origin hostname at the VPS. One secret is shared: wrangler secret put HYBRID_SHARED_SECRET, the same value the gateway holds. The Worker deploys with a single command (wrangler deploy; see hybrid/worker/DEPLOY.md). On the VPS, the gateway runs beside SpeedtestServer with environment variables naming the three bases. Clients — the operator's web page or custom engine — obtain the role-split endpoint map from GET /api/servers, a token from POST /api/sessions, measure latency against both planes, run throughput against the edge, and submit results to the VPS. The full flow is exactly what E-I executed.

The architecture is production-suitable for the operator-branded speedtest product: it is the proven speed.cloudflare.com pattern recomposed around the operator's own VPS identity, with the VPS keeping discovery, identity, UDP services and the CLI lane, and the edge supplying download and upload capacity that a single Node process — measured at 30-37% of the native daemon's single-core throughput in the baseline — could never provide. The honest caveats: throughput measured at the edge describes the client-to-PoP path, not client-to-VPS, so results must be lane-labeled (the gateway's discovery document already frames both latencies); anycast PoP selection varies between runs and should be recorded (the Worker exposes colo in __meta and response headers); the free tier's 10 ms CPU budget is tight for long sessions under load, so the paid plan is the production assumption; and upload request bodies must stay chunked below the plan cap. What this architecture cannot do is place the operator onto Ookla's public server list — listing requires the official daemon and Ookla's approval, and the CLI's closed discovery (E-o) binds that lane to the program regardless of any Workers capability.

9. Reproduction and References

All components reproduce from branch hybrid-vps-workers:

```
cd hybrid/worker && bash test/run-tests.sh  # Worker conformance, 62/62
bash hybrid/lab/harness/run-hybrid-lab.sh  # E-I hybrid session + byte accounting
bash hybrid/lab/harness/run-e-g-host-tester.sh  # E-G host-tester vs Worker, 10/10
python3 hybrid/lab/harness/e-a-cf-edge-probe.py  # E-A real-edge probes
```

The E-D and E-E official-CLI experiments use the hybrid configuration proxy (per-entry server-list rewrite, live fetch, non-publishing guarantee) via hybrid/lab/harness/run-cli-experiment.sh, and consume live configuration-endpoint quota. Primary repository references: hybrid/reports/HYBRID_RESEARCH_REPORT.md, hybrid/reports/WORKERS_LIMITS.md (claims register and deviations), validation/PROTOCOL_CONTRACT.md, and validation/diversion-lab/. External platform references (Workers limits and TCP-sockets docs, Spectrum configuration, the inbound-TCP private-beta announcement) were verified live on 2026-11-06.